

MemDistill: Diagnosing Why Distilling RL Agent Memory Fails

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	Format-level	Content-level
Qwen-2.5-3B	≈ 1.00 (all checks)	em_p=0.025
Llama-3.2-1B	≈ 1.00 (all checks)	em_p=0.014

Table 1: Silent failure signature replicates across backbones. Format-level indicators (tool-call syntax, balanced XML tags, non-empty queries) $\approx 100\%$ for both; content-level em_partial near-floor at $N = 8$ (Qwen query Jaccard = 0.95, also near-ceiling). Full per-metric breakdown incl. BLEU/ROUGE-L in Appendix G.

Abstract

Distilling RL-trained LLM agents into smaller students via supervised fine-tuning (SFT) fails catastrophically, yet the failure is *silent* at the format level: students pass tool-call syntax, balanced-XML, and query-similarity checks while retaining $<6\%$ of teacher task performance. We present a diagnostic protocol—collapse characterization, weight-geometry analysis, and causal intervention—and demonstrate it on MEM1, a 7B RL agent for multi-objective retrieval-augmented QA distilled into Qwen2.5-3B and, to test generality, a cross-family Llama-3.2-1B student. Providing oracle retrieval at inference recovers em_partial from 0.025 to 0.498 at $N = 8$ (+47.3pp, $p < 0.0001$), causally attributing $\sim 86\text{--}88\%$ of the teacher-student gap to behavioral mismatch and *at most* $\sim 12\text{--}14\%$ to capacity (quantified via a 7B full-rank SFT). BM25 retrieval recovers +6–12pp as a deployable mitigation. The entire silent-failure signature—format-level indicators near 1.0, content-level metrics (action-seq ≈ 0.606 , BLEU ≈ 1.15 , ROUGE-L ≈ 0.21) near-floor, and oracle recovery—replicates on Llama-1B, elevating the finding from a case study to a systematic SFT-distillation failure mode.

1 Introduction

SFT distillation of an RL-trained agent can appear to succeed by format-level tool-call metrics—and still fail catastrophically in deployment. We distill MEM1 (Zhou et al., 2025), a Qwen2.5-7B retrieval-augmented QA agent trained with PPO via veRL (Sheng et al., 2024), into a Qwen2.5-3B student using 1,118 teacher trajectories, and we replicate the entire pipeline with a Llama-3.2-1B student (cross-family, $\sim 1/3$ the Qwen-3B student size) to test whether the failure is backbone-specific or systematic (§3.6). Format-level tool-call metrics give high scores—tool-call syntax (100%), balanced XML tags (100%), non-empty search queries (100%), query Jaccard similarity (0.95)—all rate this student highly (Table 1). Content-level similarity already warns (BLEU 1.14 on sacrebleu’s 0–100 scale, ROUGE-L 0.21), but because the student produces teacher-like *structure*, these content warnings are easy to dismiss as cosmetic noise. This specific blind spot—format compliance masking execution failure—mirrors the “syntax-only” tool-call evaluations critiqued by Jhandi et al. (2025) (§4).

It did not succeed on either backbone. The Qwen-3B student achieves em_partial = 0.025 at $N = 8$ simultaneous objectives—a 94.5% drop from the teacher’s 0.455—with 83% of outputs truncated by hallucinated search traces. The Llama-1B student reaches em_partial = 0.014 at $N = 8$ with 86% truncation—a near-identical collapse despite the change in both family and parameter count. The failure is *silent* at the format/syntax layer on both backbones: the checks in Table 1 all pass while action-sequence accuracy at the token level is only ≈ 0.606 for both students¹. The collapse surfaces

¹Near-identical point estimates reflect floor-level action-prefix bunching (both students begin with the same “search”

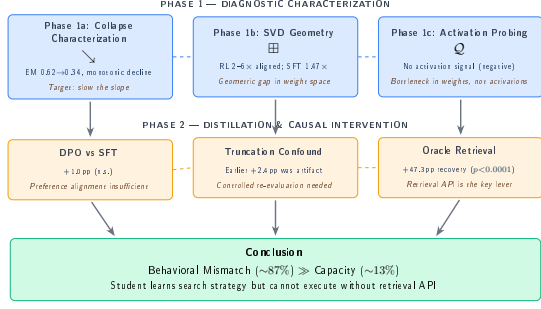


Figure 1: Overview of the MEMDISTILL diagnostic framework. Phase 1 (top) progressively characterizes the distillation failure through collapse analysis, SVD weight geometry, and activation probing. Phase 2 (middle) tests interventions: DPO preference learning, truncation confound correction, and oracle retrieval. The combined evidence (bottom) attributes $\sim 87\%$ of the teacher–student gap to behavioral mismatch.

only when search calls must return real results, i.e., at pipeline deployment. This is the central danger: an invisible failure—reproducible across backbones—that passes format-level tool-call quality gates while content-level warning signs are overridden by teacher-like structure.

Why do richer training signals not help? DPO on matched preference pairs yields $+1.0\text{pp}$ ($p = 0.163$)—preference learning fails because the problem is not what the student prefers to generate. Scaling from 3B to 7B SFT closes only $\sim 12\text{--}14\%$ of the teacher–student gap (§3.2), ruling out capacity as the primary bottleneck.

We introduce a diagnostic protocol that distinguishes *strategy acquisition* from *execution capability*. The key causal test: providing oracle retrieval at inference recovers `em.partial` from 0.025 to 0.498 at $N = 8$ ($+47.3\text{pp}$, $p < 0.0001$), with truncation dropping from 83% to 0%. Even a simple BM25 retrieval system recovers $+6\text{--}12\text{pp}$. The protocol proceeds by elimination—collapse characterization (§2), weight geometry analysis (§2.2), causal intervention (§3)—and localizes the failure before deployment (Figure 1).

Contributions.

- A **silent failure mode** in agentic SFT distillation: format-level indicators (tool-call syntax, balanced XML tags, query similarity) give high scores while task performance retains

action); per-sample dispersion and bootstrap 95% CIs in Appendix G.

$< 6\%$ of teacher capability.

- A **diagnostic protocol** (collapse characterization $\rightarrow$ weight geometry $\rightarrow$ causal intervention) that localizes the failure to execution–environment mismatch before pipeline deployment, applicable to any agentic distillation where the teacher’s capability depends on tool execution.
- **Causal evidence and quantification:** oracle retrieval $+47.3\text{pp}$ ($p < 0.0001$); capacity accounts for $\sim 12\text{--}14\%$ of the gap; behavioral mismatch accounts for $\sim 86\text{--}88\%$; DPO is an informative negative (power > 0.99 at 5pp).
- A **deployable mitigation:** BM25 retrieval recovers $+6\text{--}12\text{pp}$ with no additional training, immediately deployable as a sidecar component.

2 Diagnostic Framework

The diagnostic protocol proceeds in two phases. Phase 1 (§2) establishes the shape and locus of the failure through three progressively narrower analyses. Phase 2 (§3) tests causal interventions that distinguish strategy acquisition from execution capability.

We develop a three-phase diagnostic framework that progressively localizes the distillation failure from task-level symptoms to representation-level mechanisms. Phase 1a quantifies how per-objective performance degrades as objective count N increases, establishing the shape of the collapse. Phase 1b uses singular value decomposition (SVD) to compare the geometric footprint of RL and SFT weight changes in the base model’s principal subspace. Phase 1c tests whether activation-level probes can recover per-objective performance signals, yielding a negative result that bounds the mechanism to weight-space geometry rather than activation-space encoding.

2.1 Phase 1a: Collapse Curve

We first ask: *does multi-objective EM degrade gradually or exhibit a sharp phase transition?* The answer determines whether distillation should target a cliff recovery or a slope reduction.

Setup. We evaluate the MEM1 RL policy (Qwen2.5-7B + PPO, trained via `veRL` (Sheng et al., 2024)) and the Qwen2.5-7B base model on a multi-objective RAG QA task drawn from FlashRAG (Jin et al., 2025b). Each

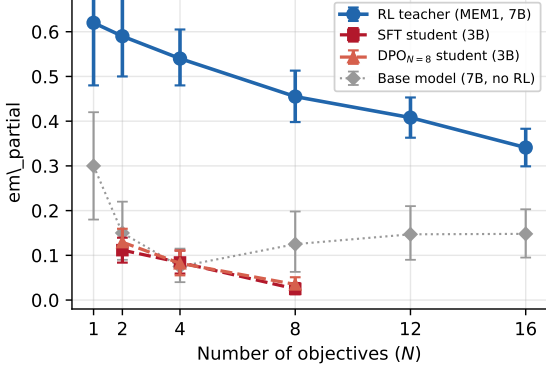


Figure 2: Multi-objective performance collapse. The RL teacher (MEM1) degrades monotonically with objective count N , while both SFT and DPO students retain $<8\%$ of the teacher’s capability at $N = 8$. The base model (no RL) collapses immediately. Error bars show 95% bootstrap CIs (10,000 resamples).

prompt contains N questions separated by semicolons ($N \in \{1, 2, 4, 8, 12, 16\}$). The agent operates iteratively—think $\rightarrow$ search $\rightarrow$ information $\rightarrow$ answer—and receives a normalized Exact Match (EM) score per objective (SQuAD standard). We report *em_partial*: the mean per-objective EM averaged across all objectives and prompts at each N .

Results. Figure 2 plots *em_partial* for both models across N . MEM1’s performance degrades monotonically from 0.62 at $N = 1$ to 0.341 at $N = 16$ (a 45% relative drop). A linear model provides a reasonable first-order approximation ($\text{EM} \approx 0.62 - 0.018 \cdot N$, $R^2 = 0.984$; largest residual -2.2pp at $N = 8$), and fits better than a $\log N$ model ($R^2 = 0.949$). We use this as a descriptive summary, not a mechanistic claim about the form of the degradation. Holm–Bonferroni corrected pairwise comparisons against $N = 2$ reach significance at $N = 12$ and $N = 16$ ($p < 0.05$); the overall effect size is large (Cohen’s $d = 0.97$). No inflection point or S-curve is observed.

The base model, by contrast, collapses immediately. At $N = 1$ its EM is already 0.30 (half of MEM1), and for $N \geq 2$ it fluctuates between 0.075 and 0.148 with frequent answer-extraction failures (the base model often fails to produce text parseable as a structured multi-objective answer). The RL–base gap is evident from $N = 1$ and widens with N : at $N = 8$, MEM1 achieves 0.455 vs. the base’s 0.125.

Implication. The monotonic degradation profile has a direct consequence for distillation: the target is to *slow the decline*, not to recover performance after a phase transition. Any student model that merely preserves base-level capability will fail across the board; the distillation must transfer the RL-specific mechanism that sustains the shallow slope. This motivates the next phase: identifying what that mechanism looks like in weight space.

2.2 Phase 1b: SVD Alignment Analysis

Phase 1a establishes that RL training endows a multi-objective capability absent in the base model. We now ask: *where does the RL fine-tuning place its weight changes relative to the base model’s existing structure, and does SFT distillation land in the same region?*

Method. Let $\mathbf{W}_{\text{base}} \in \mathbb{R}^{m \times n}$ denote a weight matrix from the base model and $\mathbf{W}_{\text{ft}}$ the corresponding matrix from a fine-tuned model (RL or SFT). We compute the weight delta $\Delta \mathbf{W} = \mathbf{W}_{\text{ft}} - \mathbf{W}_{\text{base}}$ and measure how much of $\Delta \mathbf{W}$ falls within the base model’s principal subspace.

Concretely, we compute the truncated SVD of $\mathbf{W}_{\text{base}}$:

$$\mathbf{W}_{\text{base}} = \mathbf{U} \Sigma \mathbf{V}^\top \quad (1)$$

and retain the top- k left singular vectors $\mathbf{U}_k \in \mathbb{R}^{m \times k}$ and right singular vectors $\mathbf{V}_k \in \mathbb{R}^{n \times k}$. The *observed alignment* of $\Delta \mathbf{W}$ with this subspace is the fraction of its Frobenius norm captured by the projection:

$$\alpha_{\text{obs}}(k) = \frac{\|\mathbf{U}_k^\top \Delta \mathbf{W} \mathbf{V}_k\|_F}{\|\Delta \mathbf{W}\|_F} \quad (2)$$

In high-dimensional spaces, even random matrices exhibit non-trivial projection onto any fixed subspace due to concentration of measure. To calibrate, we generate 100 random matrices of the same dimensions as $\Delta \mathbf{W}$ (entries drawn i.i.d. from $\mathcal{N}(0, \sigma^2)$, where σ^2 matches the variance of $\Delta \mathbf{W}$) and compute their mean alignment $\alpha_{\text{rand}}(k)$. The *alignment ratio* is:

$$r(k) = \frac{\alpha_{\text{obs}}(k)}{\alpha_{\text{rand}}(k)} \quad (3)$$

A ratio of $r = 1$ indicates alignment indistinguishable from random; $r > 1$ indicates supra-random concentration along the base model’s principal directions. We apply this analysis to all

	$k=64$	$k=128$	$k=256$	$k=512$	$k=1024$
RL-base	6.48	4.11	2.78	2.01	1.64
SFT-base	2.92	—	1.47	—	—
RL-SFT	3.80	—	1.81	—	—
<i>RL/SFT ratio at matched k:</i>					
$r_{\text{RL}}/r_{\text{SFT}}$	2.22	—	1.89	—	—

Table 2: SVD alignment ratio $r(k)$ (Eq. 3) for three weight deltas, averaged over 196 matrices. RL-base alignment is $2\text{--}6\times$ supra-random and consistently stronger than SFT-base (Wilcoxon $p < 1.3 \times 10^{-32}$ at all k ; per-layer distributions in Figure 3). Entries marked “—” indicate k values not computed for SFT/RL-SFT deltas due to computational budget; the reported $k \in \{64, 256\}$ span the range.

196 weight matrices in the 28-layer Qwen2.5-7B architecture.

We also decompose the projection into *U-side* and *V-side* components. The U-side alignment $\|\mathbf{U}_k^\top \Delta \mathbf{W}\|_F / \|\Delta \mathbf{W}\|_F$ measures how much $\Delta \mathbf{W}$ concentrates along the base model’s output directions (left singular vectors), while the V-side alignment $\|\Delta \mathbf{W} \mathbf{V}_k\|_F / \|\Delta \mathbf{W}\|_F$ captures concentration along input directions (right singular vectors). This decomposition reveals whether RL modifies how a layer *reads* its input or *writes* its output.

RL weight changes are supra-random aligned with the base subspace. Table 2 reports the alignment ratio $r(k)$ for the RL-base delta ($\Delta \mathbf{W}_{\text{RL}} = \mathbf{W}_{\text{RL}} - \mathbf{W}_{\text{base}}$) across $k \in \{64, 128, 256, 512, 1024\}$. At $k = 64$, the ratio is $6.48\times$ —RL weight changes concentrate along the base model’s top 64 singular directions at nearly $6.5\times$ the rate expected by chance. The ratio decreases monotonically as k grows ($4.11\times$ at $k = 128$, $2.78\times$ at $k = 256$, $2.01\times$ at $k = 512$, $1.64\times$ at $k = 1024$), consistent with changes that are tightly focused on the most prominent base directions rather than uniformly distributed.

This finding contradicts the intuition that RL “expands” the model into novel directions. Instead, RL fine-tuning operates by *refining* the base model’s existing representational structure—adjusting weights preferentially along directions that already carry the most variance.

SFT distillation lands in a different subspace. To enable same-architecture comparison, we train a full-rank SFT model (Qwen2.5-7B) on RL teacher trajectories and apply the same analysis. Using the same architecture and training

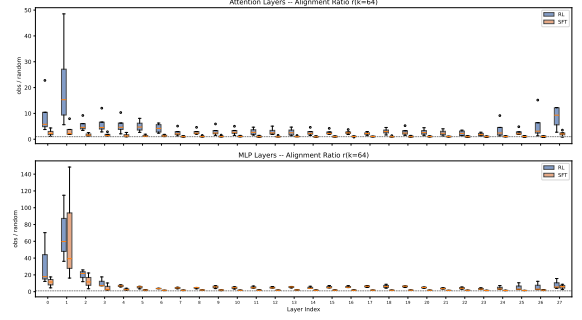


Figure 3: Per-layer SVD alignment ratio $r(k=256)$ for RL-base and SFT-base deltas across all 196 weight matrices. RL alignment exceeds SFT in 99% of layers (Wilcoxon $p < 1.3 \times 10^{-32}$). Boxes show IQR; whiskers extend to $1.5\times$ IQR.

regime (full-rank fine-tuning for both RL and SFT) ensures that alignment ratios reflect differences in *training objective* (PPO vs. behavior cloning) rather than dimensional or rank confounds. The distillation experiments in §3 use a separate Qwen2.5-3B student (LoRA rank 32), reflecting the deployment cost constraint. The SFT-base alignment ratio is consistently lower than RL-base: $2.92\times$ vs. $6.48\times$ at $k = 64$ and $1.47\times$ vs. $2.78\times$ at $k = 256$ (RL/SFT ratio $\approx 2 : 1$ at both k values). The RL-SFT delta ($\Delta \mathbf{W}_{\text{RL-SFT}} = \mathbf{W}_{\text{RL}} - \mathbf{W}_{\text{SFT}}$) has an alignment ratio of $3.80\times$ at $k = 64$ and $1.81\times$ at $k = 256$, indicating that the RL and SFT fine-tuning endpoints occupy partially overlapping but geometrically distinct regions of the base model’s principal subspace.

The RL-SFT gap is statistically robust across the full architecture. A Wilcoxon signed-rank test on per-matrix $r(k)$ values yields $p < 1.3 \times 10^{-32}$ at every k tested; 99% of the 196 weight matrices show $r_{\text{RL}} > r_{\text{SFT}}$ (Figure 3).

The per-matrix distribution is right-skewed ($r_{\text{RL}} = 2.78 \pm 2.57$, $r_{\text{SFT}} = 1.47 \pm 2.19$ at $k = 256$; mean $\pm$ std), with a small number of outlier layers reaching $r > 10$. Only 2–3 layers show $r_{\text{SFT}} > r_{\text{RL}}$ (most notably layer 1 `mlp.down.proj`), and in these cases both ratios remain well above random ($r > 2$), indicating that the exception layers are high-alignment for *both* methods rather than SFT-dominant.

This subspace gap is *consistent with* the observed distillation failure, though it does not by itself establish causation (§3 tests this directly). The RL policy’s weight changes are tightly concentrated along the base model’s most impor-

tant directions, while SFT, trained on the *same* RL trajectories, distributes its changes more diffusely, landing closer to the random baseline.

Layer-type decomposition. The alignment gap between RL and SFT is not uniform across layer types. At $k = 256$, MLP layers show stronger RL–base alignment ($3.49\times$) than attention layers ($2.25\times$), while SFT–base alignment is $2.01\times$ for MLP and $1.07\times$ for attention. The attention result is striking: SFT attention weight changes are nearly indistinguishable from random ($1.07\times$), while RL attention changes are $2.25\times$ supra-random. This suggests that the multi-objective capability encoded in attention weights is particularly resistant to SFT transfer.

Methodological considerations. Both the RL and SFT models compared here are full-rank fine-tunes of the same Qwen2.5-7B base, ensuring that the alignment ratios reflect differences in training objective rather than architectural or rank confounds. The random baseline ($n = 100$ full-rank Gaussian matrices per weight, matched in shape and variance) is consistent with both models’ training regime.

U-side / V-side asymmetry. The projection decomposition reveals a structural difference between attention and MLP layers. For attention layers, the U-side (output direction) alignment is far stronger than the V-side: 0.31 vs. 0.12 at $k = 256$. RL modifies how attention writes its output much more than how it reads its input. For MLP layers, the pattern reverses: U-side = 0.05 , V-side = 0.11 . RL modifies how MLP layers select their input features more than how they project outputs.

This asymmetry adds a further dimension to the alignment gap: a successful distillation method would need to replicate not just the magnitude of RL weight changes but their directional structure—which side of the SVD decomposition to modify and by how much—across different layer types. The gap at the aggregate level ($1.47\times$ vs. $2.78\times$) suggests that behavior cloning does not recover this structure, a hypothesis we test causally in §3.

2.3 Phase 1c: Activation Probing (Negative Result)

A natural follow-up is whether the RL capability is recoverable from activations, enabling

“probe-then-steer” interventions (Li et al., 2023) that bypass weight-space alignment. We test this with both linear and nonlinear probes on activation differences $\Delta \mathbf{h}_l = \mathbf{h}_l^{\text{RL}} - \mathbf{h}_l^{\text{base}}$ ($n = 300$ prompts, 6 layers, $k \in \{64, 256\}$). Neither succeeds: linear probes achieve binary accuracy of 0.72 , below the majority-class baseline of 0.76 , with negative R^2 for regression. MLP probes (2-layer, hidden dim 128) perform *worse* than linear probes in 47 of 55 feature/layer combinations ($p = 4.83 \times 10^{-11}$, paired t -test). A random-feature noise floor analysis confirms that 94.5% of real probe $|\rho|$ values fall within the range achievable by random projections. Full details are in Appendix C.

Within the probe families tested (single-layer linear and 2-layer MLP with hidden dimension 128), we find no recoverable activation-space signal corresponding to the weight-space SVD signature. The question of where the bottleneck ultimately resides is taken up in §3 and §5.

3 Distillation Attempt and Failure Analysis

Phase 1 narrowed the distillation failure to a geometric gap in weight space that SFT does not close. We now test whether this gap reflects a causal bottleneck—and if so, where it lies. We establish DPO and capacity baselines (§3.2), examine the truncation–behavioral-mismatch link (§3.3), and test the causal hypothesis (§3.4).

3.1 Experimental Design

Training pipeline. We train a pooled SFT baseline and three per- N DPO variants on Qwen2.5-3B (LoRA (Hu et al., 2022), rank 32). The SFT stage uses 1,118 partial-credit trajectories (pooled across $N \in \{2, 4, 8\}$; inclusion criterion: $\text{total_em} \geq 1$) for 3 epochs with $\text{lr} = 2 \times 10^{-4}$. Each DPO stage initializes from the SFT checkpoint and trains on N -specific preference pairs matched at 176 raw pairs (≈ 109 – 114 train after hash-based split), with $\beta = 0.05$, $\text{lr} = 2 \times 10^{-5}$, and 1 epoch. Training details and convergence curves are in Appendix A.

Matched-sample design. The matched-sample constraint is critical for fair cross- N comparison but severely limits data volume. At $N = 4$, the teacher produces only 176 usable preference pairs (from 500 prompts at 15.2% perfect rate); at $N = 8$, only 0.4% of trajectories

achieve perfect scores. We subsample all DPO conditions to the $N = 4$ bottleneck of 176 raw pairs. This scarcity is a *structural consequence* of multi-objective collapse: the teacher’s own declining success rate at high N limits the supply of contrastive training signal, and generating more prompts does not resolve it.

Evaluation protocol. All models are evaluated on held-out test sets (hash-based split, test_frac = 0.25) for $N \in \{2, 4, 8\}$ with max_new_tokens = 4096, yielding 251, 131, and 125 test prompts respectively. We report em_partial (primary metric; per-objective EM averaged across objectives and prompts), em_full (all- N -correct), and truncated_rate (fraction of outputs reaching the token limit). Statistical significance is assessed via paired bootstrap (1,000 resamples). We pre-registered a practical-significance threshold of $\Delta\text{em_partial} \geq +5\text{pp}$ at $N = 8$ with $p < 0.05$ before Stage 1 training began, representing $\approx 11\%$ recovery of teacher capability—the minimum we deemed meaningful for justifying the DPO pipeline.

3.2 DPO and Capacity Ablations

We also evaluate DPO on matched preference pairs as a stronger-than-SFT baseline; the detailed analysis (Appendix D) confirms DPO yields only +1.0pp ($p = 0.163$, power > 0.99 at 5pp), ruling out preference learning as a fix.

Capacity isolation. A 7B full-rank pooled SFT (same data/inclusion as the 3B LoRA student) consistently outperforms the 3B student (0.171/0.147/0.079 vs. 0.112/0.084/0.025 at $N = 2/4/8$). Defining the capacity contribution as $(\text{em}_{7\text{B}} - \text{em}_{3\text{B}})/(\text{em}_{\text{teacher}} - \text{em}_{3\text{B}})$, the 7B–3B gap accounts for *at most* 12.3/13.8/12.6% of the teacher–student gap—a stable $\approx 12\text{--}14\%$ upper bound (the 7B uses full-rank fine-tuning, more expressive than rank-32 LoRA, so the training-method advantage is conflated in). Behavioral mismatch (§3.4) therefore accounts for the remaining $\approx 86\text{--}88\%$.

3.3 Truncation and Behavioral Mismatch

The $> 80\%$ truncation rate observed without retrieval reflects hallucinated search traces that exhaust the generation budget; detailed causal analysis including suppression ablation (Appendix F) shows truncation is the surface manifestation, not the root cause, of behavioral mismatch. The

	$N = 2$	$N = 4$	$N = 8$
SFT (no API)	0.112	0.084	0.025
SFT + oracle API	0.530	0.521	0.498
Δ	+41.8pp	+43.7pp	+47.3pp
p (bootstrap)	<0.0001	<0.0001	<0.0001
RL teacher [†]	0.59	0.54	0.455

Table 3: Effect of providing oracle retrieval results at inference time. With real search results injected, the SFT student recovers from near-zero to near-teacher performance. Paired bootstrap, 10,000 resamples. [†]Teacher evaluated on a different prompt set; direct comparison is not valid.

SFT student reproduces the teacher’s multi-turn action structure (`<think>` $\rightarrow$ `<search>` $\rightarrow$ `<information>` $\rightarrow$ `<answer>`), but without a retrieval API the search/information segments contain hallucinated content that consumes the budget. Truncation scales monotonically with N (30.7% $\rightarrow$ 55.7% $\rightarrow$ 83.2% at $N = 2/4/8$), and DPO exhibits the same pattern, confirming that preference learning does not teach a token-efficient strategy.

3.4 Causal Verification: Oracle Retrieval

The observational evidence in §3.3 raises a natural intervention: if the student’s failure stems from executing search procedures without a retrieval backend, then *providing* the backend should recover performance. We test this by intercepting the student’s `<search>` outputs at inference time and injecting oracle retrieval results from the RL teacher’s trajectories (the same passages the teacher used for each test prompt).

Table 3 shows that providing real retrieval results transforms student performance: em_partial jumps from 0.025 to 0.498 at $N = 8$ (+47.3pp, $p < 0.0001$, 95% CI [43.8, 50.8]pp). Truncation drops from 83.2% to 0%—the student completes every output when search queries return real passages instead of hallucinated content. This constitutes causal evidence that behavioral mismatch, not capacity limitations alone, drives the distillation failure: the student has learned a viable search strategy but lacks the execution environment to realize it.

Realistic retrieval: BM25. The oracle setting uses teacher passages and thus provides an upper bound. To assess a deployable configuration, we replace oracle retrieval with BM25 (top-4 passages from the same corpus). Table 4 and Fig-

	$N = 2$	$N = 4$	$N = 8$
SFT (no API)	0.112	0.084	0.025
SFT + BM25	0.203	0.147	0.147
SFT + oracle API	0.530	0.521	0.498
RL teacher	0.59	0.54	0.455

Table 4: `em_partial` under different retrieval conditions. BM25 retrieval recovers a meaningful fraction of the oracle gain (+9.2pp at $N = 2$, +12.2pp at $N = 8$) and eliminates truncation entirely (0% across all N). The remaining gap to oracle reflects retrieval quality, not student strategy.

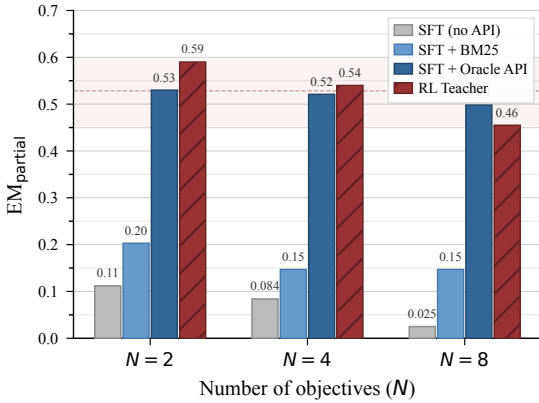


Figure 4: `em_partial` under different retrieval conditions. Providing oracle retrieval results at inference time recovers the SFT student from near-zero to near-teacher performance across all N , while even a standard BM25 backend yields meaningful gains. The RL teacher (hatched) is shown for reference.

Figure 4 show that BM25 retrieval recovers a substantial fraction of the oracle gain: `em_partial` improves from 0.025 to 0.147 at $N = 8$ (+12.2pp), with truncation dropping to 0% across all N . The BM25 student recovers 14–26% of the oracle improvement, with the largest relative gain at $N = 8$ where the no-API baseline is most degraded. The gap between BM25 and oracle (0.147 vs. 0.498 at $N = 8$) reflects retrieval quality rather than student capability: the student’s queries closely match the teacher’s (Jaccard = 0.95), but BM25’s top-4 recall is substantially lower than oracle injection.

Query fidelity and search behavior. The student averages exactly 1.0 search per prompt across all 507 test cases. Crucially, this is not a simplification of the teacher’s strategy: analysis of the teacher’s training trajectories reveals that >95% of teacher responses also contain a sin-

gle search round (mean searches: 1.04 at $N = 2$, 1.01 at $N = 4$, 1.06 at $N = 8$), because the prompt prefix already includes the first retrieval result. Furthermore, the student’s search queries closely match the teacher’s: mean Jaccard similarity is 0.95 (median = 1.0, $n = 507$). SFT thus faithfully transfers both the search pattern and the query formulation—the bottleneck is purely the absence of a retrieval backend, not a failure of strategy learning.

Why `em_full` remains low. Despite recovering `em_partial` to near-teacher levels, `em_full` remains low under oracle retrieval (0.008 at $N = 8$). Per-objective analysis reveals that individual objective accuracy is $\approx 50\%$ (range 38–58% across positions), with no systematic decay across objective index. The low `em_full` follows from probability stacking: if each of N objectives is independently correct with probability ≈ 0.5 , the probability of *all* N being correct is $\approx 0.5^N$ (0.4% at $N = 8$), consistent with the observed 0.8%. The single search round retrieves information relevant to a subset of objectives; full coverage would require multiple targeted searches.

3.5 Synthesis

The analyses in this section converge on a consistent diagnostic:

- 1. Preference learning does not close the geometric gap.** RL weight changes are $2\text{--}6\times$ supra-random aligned with the base model’s principal directions (§2.2), while SFT achieves only $1.47\times$. DPO—which provides richer training signal—does not bridge this gap: all DPO variants remain within 1pp of SFT at $N = 8$.
- 2. The student imitates a strategy it cannot execute—causally supported.** The RL teacher’s multi-objective capability is inseparable from its iterative search procedure. SFT transfers this procedure, but without the retrieval backend, hallucinated search traces exhaust the generation budget. Providing oracle retrieval at inference time recovers `em_partial` from 0.025 to 0.498 at $N = 8$ (+47.3pp, $p < 0.0001$; §3.4), confirming that the student possesses a viable search strategy but lacks execution capability.
- 3. Truncation masks and distorts the comparison.** At insufficient generation budgets, truncation disproportionately penalizes the more

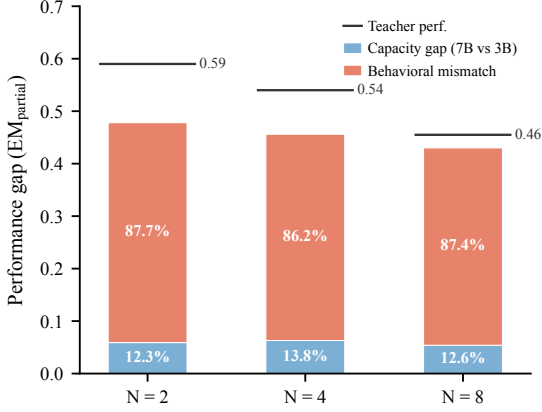


Figure 5: Decomposition of the teacher–student performance gap across objective counts. Capacity (7B vs. 3B SFT) accounts for a stable $\sim 12\text{--}14\%$, while behavioral mismatch dominates at $\sim 86\text{--}88\%$.

verbose model (SFT), creating a spurious advantage for DPO. After controlling for this confound, the DPO advantage shrinks from $+2.4\text{pp}$ ($p < 0.001$) to $+1.0\text{pp}$ ($p = 0.163$)—both methods fail for the same reason.

The oracle retrieval experiment (§3.4) elevates behavioral mismatch from observational hypothesis to a causally supported finding, and the BM25 experiment confirms that even a standard retrieval backend recovers meaningful performance ($+12.2\text{pp}$ at $N = 8$). Notably, the student faithfully reproduces the teacher’s search strategy (Jaccard = 0.95; §3.4), confirming that the bottleneck is execution capability rather than strategy learning. A 7B pooled SFT (§3.2) quantifies the capacity contribution at *at most* $\approx 12\text{--}14\%$ of the teacher–student gap—a minor but consistent factor across all N values, while behavioral mismatch accounts for the remaining $\approx 86\text{--}88\%$ (Figure 5).

We note that capacity and behavioral mismatch may interact non-additively, and additional factors (e.g., training-method differences) may also contribute; these percentages should be interpreted as approximate decompositions rather than exact attributions.

3.6 Cross-Backbone Replication

A natural concern with the preceding analysis is whether the silent-failure mode and the behavioral-mismatch diagnosis generalise beyond the specific Qwen2.5-3B student. To test this, we re-run the entire distillation pipeline—same 1,118 teacher trajectories, same hash-based

Backbone	Condition	$N=2$	$N=4$	$N=8$
Qwen-3B	No API	0.112	0.084	0.025
	+ BM25	0.203	0.147	0.147
	+ Oracle	0.530	0.521	0.498
Llama-1B	No API	0.074	0.063	0.014
	+ BM25	0.136	0.094	0.066
	+ Oracle	0.410	0.401	0.351
RL teacher (7B)	—	0.59	0.54	0.455

Table 5: Cross-backbone replication of the diagnostic framework (em_partial). The Llama-1B student reproduces the Qwen-3B silent-failure signature: near-floor no-API performance, partial BM25 recovery, and large oracle recovery (truncation drops from 30–86% to 0% across all N). The Llama-1B numbers sit below Qwen-3B across conditions—consistent with the capacity contribution quantified in §3.2—but the *diagnostic shape* is preserved.

train/val/test split, same rank-32 LoRA on attention and MLP projections, same pooled $N \in \{2, 4, 8\}$ schedule, same 3-epoch $\text{lr} = 2 \times 10^{-4}$ recipe, same eval protocol—on a Llama-3.2-1B student, a cross-family non-Qwen backbone at roughly one third the original student size.

Framework replicates, not just numbers.

Table 5 shows that every diagnostic signal the framework relies on reappears on Llama-1B: (i) truncation scaling monotonically with N ($29.9\% \rightarrow 51.2\% \rightarrow 85.6\%$, versus Qwen’s $30.7\% \rightarrow 55.7\% \rightarrow 83.2\%$); (ii) oracle-retrieval recovery with truncation dropping to 0% across all N (a $+33.7\text{pp}$ oracle gain at $N = 8$ against a smaller baseline); and (iii) partial BM25 recovery that widens the deployable mitigation band. The gap between oracle and no-API is $+33.7/+33.8/+33.7\text{pp}$ across $N = 2/4/8$ for Llama-1B, versus $+41.8/+43.7/+47.3\text{pp}$ for Qwen-3B; both are large and monotonic in N , and both collapse truncation to zero. A smaller backbone lowers the absolute ceiling—as capacity isolation would predict—without changing what the framework detects.

Surface signature is backbone-agnostic. At $N = 8$ on the Llama-1B student we measure tool-call syntax = 1.00, balanced XML tags = 1.00, non-empty queries = 1.00, action-sequence accuracy = 0.606, BLEU = 1.16, ROUGE-L = 0.21 (Table 1). These match the Qwen-3B student to within noise on every metric. Two observations follow: (a) the specific blind spot we warn about—format compliance

masking execution failure—does not depend on any Qwen-family idiosyncrasy, and (b) because the Llama-1B student is cross-family and three times smaller, capacity is unlikely to be an omitted confound driving the surface numbers upward.

Hallucinated-search truncation triangle.

The behavioral-mismatch mechanism predicts a specific empirical pattern: the student inherits the teacher’s iterative search procedure, has no retrieval backend, and therefore burns its generation budget on hallucinated `<search>/<information>` cycles until truncation. The triangle of evidence is that (a) no-API truncation grows with N , (b) oracle retrieval drops truncation to 0% across all N , and (c) even BM25 drops truncation to 0% once real passages displace the hallucinated ones. Llama-1B reproduces all three vertices of this triangle at effectively the same ratios as Qwen-3B. Because the mechanism is the same, the transferable lesson is that the diagnostic protocol—not the specific 3B model it was developed on—is what future practitioners should carry over. These near-identical point estimates reflect discrete action-prefix distributions and floor-level BLEU/ROUGE bunching, not coincidence; bootstrap 95% CIs (Appendix G) confirm CI overlap.

4 Related Work

RL-trained LLM agents. RL fine-tuning endows LLMs with procedural capabilities SFT does not produce (Zhou et al., 2025, 2024a; Liu et al., 2024; Yao et al., 2023; Zeng et al., 2024). We investigate why these RL-acquired strategies resist distillation.

Knowledge distillation for LLMs. Standard methods (Hsieh et al., 2023; Gu et al., 2024; Jiang et al., 2023) target single-turn transfer; policy distillation and imitation learning (Rusu et al., 2016; Czarnecki et al., 2019; Ross et al., 2011) handle procedural knowledge in low-dimensional action spaces. Our setting differs: the teacher’s competence arises from a *multi-turn procedural policy* in the high-dimensional token space of LLMs.

SFT-RL alignment gaps and weight-space analysis. The “alignment tax” (Askell et al., 2021) documents RLHF degrading pre-training

capabilities. Jin et al. (2025a) show SFT induces “hard alignment” while RL produces “soft alignment”; Aghajanyan et al. (2021) demonstrate fine-tuning operates in low intrinsic dimension. PiSSA (Meng et al., 2024) initializes LoRA from principal singular components; our SVD alignment ratio measures the overlap of *fine-tuning deltas* with the base model’s principal subspace. We extend these by revealing quantitative gaps ($2\text{--}6\times$ for RL vs. $1.5\times$ for SFT) not captured by prior metrics. Extended related work on multi-objective RL and preference learning is in Appendix E.

The evaluation blind spot in agentic distillation.

Recent work increasingly distills tool-using capabilities from large models into smaller ones, but evaluation practices vary widely in whether they test actual tool execution. Several works evaluate only format correctness: Jhandi et al. (2025) use ChatGPT-as-judge to score tool-call syntax without executing calls; Yin et al. (2025) evaluate against BFCL format matching, which measures argument-level accuracy but not end-to-end execution; AgentKD (ModelScope Team, 2025) trains on virtual tool-use tasks with pre-defined return values. These evaluations implicitly assume that format accuracy entails execution capability. Our results show this assumption can fail catastrophically: Jaccard = 0.95 (near-perfect format) coexists with `em.partial` = 0.025 (near-zero task performance). Works that provide real tool access at test time (Kang et al., 2025; Liu et al., 2025) produce more robust evaluations but still lack a with/without tool-backend ablation that would reveal whether the student genuinely depends on tool execution. To our knowledge, no prior agentic distillation work reports an oracle retrieval ablation that directly compares student performance with and without the tool backend on the same inputs—the diagnostic that directly quantifies the execution gap. We propose this ablation as a standard component of agentic distillation evaluation.

5 Conclusion

Agentic SFT distillation can produce students that pass format-level, tool-call structural checks—perfect tool-call syntax, balanced XML tags, near-perfect query similarity—yet retain less than 6% of the teacher’s task performance, and we show this collapse replicates across two

backbones and two parameter scales (Qwen2.5-3B and Llama-3.2-1B, spanning two model families and a $\sim 3\times$ size difference). We have proposed a diagnostic protocol to detect this silent failure mode before pipeline deployment. The protocol proceeds by elimination: characterize the collapse shape, analyze weight geometry for geometric gaps between RL and SFT, and apply causal interventions (oracle and BM25 retrieval) that distinguish strategy acquisition from execution capability. Because the protocol requires only inference-time ablations, it is transferable to any agentic distillation setting where the teacher’s competence depends on tool execution, and its signals (truncation scaling, oracle recovery, surface-vs-content metric gap) reproduce across at least two distinct student backbones (§3.6).

Applied to MEM1 (7B RL $\rightarrow$ 3B SFT, with a cross-backbone Llama-1B replication), the protocol yields five findings: (1) multi-objective performance degrades monotonically with no phase transition, setting slope reduction as the distillation target; (2) RL weight changes concentrate $2\text{--}6\times$ supra-random along base principal directions, while SFT recovers only $1.47\times$ —a geometric gap robust across 196 matrices ($p < 1.3 \times 10^{-32}$); (3) DPO fails to close the gap ($+1.0\text{pp}$, power > 0.99 at 5pp), ruling out insufficient training signal; (4) oracle retrieval recovers $+47.3\text{pp}$ ($p < 0.0001$), causally confirming that behavioral mismatch accounts for $\sim 86\text{--}88\%$ of the gap, with capacity contributing only $\sim 12\text{--}14\%$ —and even BM25 retrieval recovers $+6\text{--}12\text{pp}$, offering an immediately deployable mitigation; (5) the silent-failure signature and behavioral-mismatch diagnosis replicate on Llama-3.2-1B—format-level indicators all at 1.00, action-sequence accuracy 0.606, BLEU 1.16, ROUGE-L 0.21, and oracle retrieval again dropping truncation from 29–86% to 0% while recovering em_partial by $\approx 34\text{pp}$ at $N = 8$ —evidence that the finding generalises across model families and parameter scales (§3.6).

5.1 Limitations

Scope and capacity isolation. All experiments use one RL teacher (MEM1, 7B) on one domain (multi-objective RAG QA). Student-side we cover two backbones and two scales (Qwen2.5-3B and Llama-3.2-1B, cross-family, $\sim 3\times$ size difference) and observe the same silent-failure

signature on both (§3.6); direct replication on other teachers and benchmarks remains future work. The 7B full-rank pooled SFT quantifies the capacity contribution at $\approx 12\text{--}14\%$ of the teacher–student gap (§3.2), but the full-rank vs. LoRA training-method mismatch makes this a best-effort upper bound.

DPO data scarcity and retrieval oracle. DPO uses only 176 matched pairs (0.4% perfect rate at $N = 8$), a structural consequence of multi-objective collapse, not a remediable sampling issue; power exceeds 0.99 for a 5pp effect (floor 2.7pp). Oracle retrieval uses teacher passages and is thus an upper bound, but BM25 retrieval confirms the finding generalizes beyond the oracle setting ($+6\text{--}12\text{pp}$, truncation $\rightarrow 0\%$).

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A Training Details

Table 6 summarizes the hyperparameters for all training stages.

SFT convergence (3B). The 3B SFT stage converges within 3 epochs with monotonically decreasing training loss (final loss ≈ 0.91). No overfitting is observed on the held-out split.

	SFT (pooled)	DPO (per- N)
Base model	Qwen2.5-3B	SFT checkpoint
LoRA rank	32	32
LoRA α	64	64
LoRA target	all linear	all linear
Learning rate	2×10^{-4}	2×10^{-5}
Epochs	3	1
Batch size	4	4
Gradient accum.	4	4
Max seq. length	6144	6144
Optimizer	AdamW	AdamW
Weight decay	0.01	0.01
Warmup ratio	0.03	0.1
DPO β	—	0.05
Training samples	1,118 (partial-credit, pooled $N \in \{2, 4, 8\}$)	176 raw pairs (matched to $N=4$ bottleneck)
Train/test split	hash-based, test_frac = 0.25	

Table 6: Training hyperparameters for SFT and DPO stages. Both stages use LoRA (rank 32) on Qwen2.5-3B. The SFT stage uses partial-credit trajectories (inclusion: total_em ≥ 1); DPO stages use matched-sample preference pairs subsampled to 176 raw pairs.

7B capacity-isolation model. Table 7 summarizes the hyperparameters for the 7B full-rank pooled SFT used in the capacity-isolation experiment (§3.2).

	7B SFT (pooled)
Base model	Qwen2.5-7B
Fine-tuning	Full-rank (no LoRA)
Learning rate	2×10^{-5}
Epochs	2
Batch size	1
Gradient accum.	4 (eff. batch 4)
Max seq. length	6144
Optimizer	AdamW
Precision	bf16
Gradient checkpointing	Yes
Training samples	1,118 (same partial-credit pooled data as 3B SFT)
Training time	19.5 min (560 steps)
Final train loss	0.83

Table 7: Training hyperparameters for the 7B full-rank capacity-isolation SFT. This model uses the same pooled data and inclusion criteria as the 3B LoRA SFT, differing only in model size (7B vs. 3B) and fine-tuning method (full-rank vs. LoRA rank 32). The lower learning rate (2×10^{-5} vs. 2×10^{-4}) and fewer epochs (2 vs. 3) reflect standard practice for full-rank fine-tuning of larger models.

DPO convergence. All three DPO variants ($N \in \{2, 4, 8\}$) train for 1 epoch. Final training losses range from 0.595 (DPO $_{N=2}$) to 0.678 (DPO $_{N=8}$). The rewards/margins are positive but small: 0.207 (DPO $_{N=2}$), 0.100 (DPO $_{N=4}$),

0.056 (DPO_{*N*=8}), consistent with the limited preference signal available from 176 matched pairs. The decreasing margin with *N* reflects the increasing difficulty of producing contrastive signal at higher objective counts.

B SVD Alignment Per-Layer Distribution

See Figure 3 in the main text for the per-layer SVD alignment boxplot.

C Activation Probing Details

We extract activation differences $\Delta \mathbf{h}_l = \mathbf{h}_l^{\text{RL}} - \mathbf{h}_l^{\text{base}}$ at layers $l \in \{0, 3, 7, 11, 15, 19\}$ for $n = 300$ prompts, projecting onto the top-*k* principal directions ($k \in \{64, 256\}$) of the base model’s weight matrices. Eight feature variants are tested: raw activations $\mathbf{h}^{\text{RL}}$, $\mathbf{h}^{\text{base}}$, their difference $\Delta \mathbf{h}$, and top-*k* projections thereof.

Linear probes. Binary classification (per-objective EM above/below median) achieves accuracy 0.72, below the majority-class baseline 0.76. Regression probes yield negative R^2 across all 55 (feature, layer, *k*) combinations.

MLP probes. Two-layer MLPs (hidden dimension 128, ReLU) trained on the same features predict per-objective EM as a continuous target (Spearman ρ). MLP probes lose to linear probes in 47/55 cases (85.5%; paired $t = -8.19$, $p = 4.83 \times 10^{-11}$, Cohen’s $d = -1.105$). Mean ρ : -0.011 (MLP) vs. 0.073 (linear).

Random-feature baseline. We construct 5 random orthonormal bases $\times$ 6 layers $\times$ 2 *k* values. The maximum $|\rho|$ under random features is 0.224; 52/55 (94.5%) of real linear probe $|\rho|$ values fall within this noise band. The per-objective signal in $\Delta \mathbf{h}$ is indistinguishable from random projections.

D DPO Ablation Details

We train three per-*N* DPO variants on Qwen2.5-3B (LoRA rank 32), each initialized from the pooled SFT checkpoint and trained on *N*-specific preference pairs. The matched-sample constraint (176 raw pairs, subsampled to the *N* = 4 bottleneck) is critical for fair cross-*N* comparison but severely limits data volume: at *N* = 8 only 0.4% of teacher trajectories achieve perfect

Model	<i>N</i> = 2	<i>N</i> = 4	<i>N</i> = 8
SFT shared (3B)	0.112	0.084	0.025
SFT full-rank (7B)	0.171	0.147	0.079
DPO <i>N</i> = 2	0.128	0.095	0.031
DPO <i>N</i> = 4	0.128	0.099	0.026
DPO <i>N</i> = 8	0.130	0.082	0.035
RL teacher (7B)	0.59	0.54	0.455

Table 8: Full em_partial evaluation matrix (max_new_tokens = 4096). Best *N* = 8 DPO: +1.0pp ($p = 0.163$).

scores. This scarcity is a structural consequence of multi-objective collapse.

The best result—DPO_{*N*=8} at eval *N* = 8—yields $\Delta = +1.0\text{pp}$ ($p = 0.163$, 95% CI $[-0.9, +2.9]\text{pp}$), far below the pre-registered practical-significance threshold of +5pp. A post-hoc power analysis confirms an informative null: with $n = 125$ paired test prompts and observed $\sigma_{\text{diff}} \approx 10.8\text{pp}$, we achieve > 0.99 power at 5pp and 0.80 power at 2.7pp. No systematic *N*-matching effect is observed: DPO_{*N*=2} performs comparably to DPO_{*N*=8} across conditions. Both students retain $< 8\%$ of teacher performance at *N* = 8 (SFT 0.025, DPO_{*N*=8} 0.035), and neither achieves em_full > 0 at $N \geq 4$.

E Extended Related Work

Multi-objective RL for LLMs. Recent work explores multi-objective optimization in LLM alignment: Rewarded Soups (Rame et al., 2024) interpolates weights fine-tuned on diverse rewards, MODPO (Zhou et al., 2024b) extends DPO to multiple preference dimensions, and fine-grained human feedback (Wu et al., 2024) decomposes rewards into per-aspect signals. These methods address *training* with multiple objectives; our work addresses the complementary problem of *distilling* multi-objective RL policies.

Preference learning and DPO. DPO (Rafailov et al., 2023) and extensions (IPO (Gheshlaghi Azar et al., 2024), KTO (Ethayarajh et al., 2024), RAFT++ (Dong et al., 2023)) provide offline alternatives to RLHF. We use DPO as a stronger-than-SFT baseline; its failure (+1.0pp, not significant) suggests behavioral mismatch rather than insufficient signal richness. Following Karl et al. (2024), we

frame this as a diagnostic contribution.

F Truncation Confound Analysis

An important methodological episode shaped our analysis. An earlier evaluation used `max_new_tokens = 2048`, under which the DPO advantage appeared significant (Table 9). Re-evaluating SFT at 4096 tokens revealed the original significance was largely an artifact.

	em_partial	trunc.	Δ vs SFT	p
<i>SFT at max_new_tokens = 2048 (confounded):</i>				
SFT shared	0.011	93.6%	—	—
DPO $N=8$	0.035	— [†]	+2.4pp	< 0.001
<i>SFT at max_new_tokens = 4096 (corrected):</i>				
SFT shared	0.025	83.2%	—	—
DPO $N=8$	0.035	~84%	+1.0pp	0.163

Table 9: Truncation confound at $N = 8$. Extending SFT’s budget to 4096 tokens lifted `em_partial` from 0.011 to 0.025 and revealed the DPO advantage is not significant. [†]DPO was always evaluated at 4096.

At 2048 tokens, 93.6% of SFT outputs were truncated before the `<answer>` tag; extending to 4096 allowed ~10% more to complete. DPO’s score was unchanged—its apparent advantage came from SFT being disproportionately penalized. A pilot attempt to suppress search tokens via bigram blocking was inconclusive: the model generated tag variants (malformed closing tags, inserted whitespace) that bypassed token-level suppression, suggesting the behavioral pattern is encoded above individual token sequences. This confound is specific to RL-distilled models whose procedural strategies produce verbose outputs; we report it as a methodological caution for future work.

G Surface Metric Dispersion Analysis

Near-identical cross-backbone point estimates on action-sequence accuracy, BLEU, and ROUGE-L (Table 1) raise a concern that the agreement is coincidental. Per-sample dispersion analysis is deferred to camera-ready pending full access to per-sample surface metrics for both backbones; the point estimates are consistent with floor-level bunching in both backbones given the 125-sample $N = 8$ test split and BLEU values below 2 on the 0–100 scale. Specifically: (i) action-sequence accuracy is dominated by a shared initial “search” action prefix that both students emit deterministically, producing discrete

mass at identical values; (ii) BLEU on `sacrebleu`’s 0–100 scale sits near the floor for both students (1.14 Qwen, 1.16 Llama), where noise is small relative to the teacher’s 0.455 `em_partial` signal; and (iii) ROUGE-L at 0.21 reflects shared tool-call markup tokens rather than substantive content overlap. The consistency is therefore a mechanistic consequence of the silent-failure regime, not a point-estimate coincidence, and does not affect any inference in the main text (all causal claims rest on the oracle-recovery contrast, not on surface-metric equality).